Your research, your decisions

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Assignment confirmed: Akshaj K row; semiconductors and chip equipment, plus aerospace and defence. This pack covers all 12 candidates through Round 2, as you requested. It is research support, not your submission narrative.

The method in plain language

First compare each industry with the S&P 500. Then run M12 to compare companies only with their own peers. A strong peer score does not establish fair value. DCF asks what cash the business could produce; reverse DCF asks what growth would justify today's price. Red flags and risk tests challenge the result. Your edge, competitor judgment and fit for Laura come last.

Worked DCFs below use labelled assumptions. They are examples to challenge and replace. Keep proposed rules and thresholds separate from facts. The two industry totals remain MISSING until the comparison data are made consistent.

Answer these eight questions in your own words

1. What does Laura need from us, and when? Give amounts and years.

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2. Why does each bond mature just before a payment, not after?

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3. What is the floor, and why does it let us take risk with the rest?

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4. What do VOO and VXUS do that the tilts don't? Why is XLF called a hedge?

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5. Why isn't 'a great company' enough of a reason to buy?

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6. XLI's price assumes ~16% earnings growth, XLK's ~37%. Which needs more to go right?

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7. A stock's worst year was -40%. What's the most we can put in it? Show the maths.

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8. Name two things you must never do in this research.

Questions reproduced from your team document. The XLI/XLK percentages in question 6 are its teaching assumptions, not newly verified market forecasts. Send your answers for feedback; none has been written for you.

Semiconductors and chip equipment

Industry research scorecard | Candidate order is not a ranking

Industry score: MISSING. M12 gate: 6/6 complete with proposed R8, 3/6 with R8 off. Five complete companies and 80% coverage are required. Pat must decide R8 before releasing scores.

Candidate	Country	USD cap proxy, bn	WInS eligible?
NVDA - NVIDIA	US	4,931.8	MISSING
AMD - Advanced Micro Devices	US	824.1	MISSING
AVGO - Broadcom	US	1,556.6	MISSING
QCOM - Qualcomm	US	173.0	MISSING
LRCX - Lam Research	US	298.7	MISSING
ASML - ASML	Netherlands	603.5	MISSING

Market caps use 30 October USD listing closes and dated reported shares. ASML uses EUR price and market cap in M12/DCF instead. Foreign share counts are from December 2025; these are dated proxies.

Measure	Chips	Equipment	S&P comparison
ROE	31.4%	35.8%	29.0% (Sep)
Operating margin	35.3%	26.3%	MISSING
Book debt/equity	0.34	0.47	MISSING
Expected 5y EPS growth	22.9%	18.7%	MISSING
Forward P/E	37.29	41.58	18.70 (Oct)*
P/S	15.46	7.49	MISSING
P/B	13.31	9.70	5.30 (Sep)

Industry reference: Damodaran US data, January 2026; older than the S&P references. *Forward-P/E scoring requires each industry's own 10-year average, which remains MISSING. S&P own average 19.1x does not substitute. No combined chip/equipment series is approved. The template allocates 25 points to each pillar, with three 8.3-point valuation items (99.9 total).

ETF technical proxy SOXX: USD 471.51; 200-day average 476.37; distance from 52-week high -28.1%; one-year price return 53.8%. ETF scope differs from this candidate pool.

Investigate AI infrastructure spending, equipment intensity and device diversification; test export restrictions and customer spending cycles.

Sources: industry ROE | margins / P-S | debt | growth / P-E | P-B | S&P Sep facts | S&P Oct P-E | ETF history

Your verdict: two positives, two negatives, then your top three and the reason for each.

Aerospace and defence

Industry research scorecard | Candidate order is not a ranking

Industry score: MISSING. M12 gate: 6/6 complete with proposed R8, 3/6 with R8 off. Five complete companies and 80% coverage are required. Pat must decide R8 before releasing scores.

Candidate	Country	USD cap proxy, bn	WInS eligible?
RTX - RTX	US	269.6	MISSING
LMT - Lockheed Martin	US	147.1	MISSING
GD - General Dynamics	US	100.7	MISSING
NOC - Northrop Grumman	US	79.2	MISSING
HII - Huntington Ingalls Industries	US	12.5	MISSING
ESLT - Elbit Systems	Israel	36.2	MISSING

Market caps use 30 October USD listing closes and dated reported shares. ASML uses EUR price and market cap in M12/DCF instead. Foreign share counts are from December 2025; these are dated proxies.

Measure	Aerospace/defence	S&P comparison
ROE	15.3%	29.0% (Sep)
Operating margin	8.7%	MISSING
Book debt/equity	1.21	MISSING
Expected 5y EPS growth	25.0%	MISSING
Forward P/E	45.87	18.70 (Oct)*
P/S	3.17	MISSING
P/B	7.88	5.30 (Sep)

Industry reference: Damodaran US data, January 2026; older than the S&P references. *Forward-P/E scoring requires each industry's own 10-year average, which remains MISSING. S&P own average 19.1x does not substitute. No combined chip/equipment series is approved. The template allocates 25 points to each pillar, with three 8.3-point valuation items (99.9 total).

ETF technical proxy ITA: USD 209.74; 200-day average 228.93; distance from 52-week high -18.3%; one-year price return -2.8%. ETF scope differs from this candidate pool.

Investigate funded backlog, aftermarket demand and production capacity; test fixed-price overruns, supply constraints and budget timing.

Sources: industry ROE | margins / P-S | debt | growth / P-E | P-B | S&P Sep facts | S&P Oct P-E | ETF history

Your verdict: two positives, two negatives, then your top three and the reason for each.

NVDA | NVIDIA

Semiconductors and chip equipment | One-page worksheet | All valuations below are illustrative

US | Nasdaq | USD earnings, US GAAP | FY0 ends 2026-01-25. Price USD 204.64 (30 Oct); market-cap proxy USD 4,931.8bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	24.35x / 10.42x / 13.23x
Value per share; bear at beta 1	USD 4,983.52; bear 184.82 (0.90x price); upside/downside 241.06
Reverse DCF; SBC	First-year growth NOT DEFINED in search bracket; reported SBC/revenue 3.0%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: 4; M: -1.17. Screening flags are not proof of manipulation.

Example assumptions: sales CAGR 100.0% repeated for five years, then fade to 3%; GAAP margin flat; tax 15.1%; WACC beta 1 9.9%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Critical example limit: 100% historical growth extrapolated for five years produces an extreme value. It is not a supported forecast or an edge.

Risk: beta 2.21; volatility 49.9%; Sharpe 1.01; Treynor 0.23; alpha 34.7%. Worst price calendar year 2002: -82.8%; $6,000/loss = USD 7,247. Adjusted-return cap: USD 7,247. Historical caps are not recommended allocations.

Technical: -15.9% from 52-week high; USD price 204.64 vs 200-day mean 204.38; 1y price return 1.1%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): yes, 8.4%.

Guidance (2026-08-26): Q3 FY2027 revenue USD 108,000m +/-2%; GAAP and non-GAAP gross margin 74.0% +/-0.5 percentage points. The revenue outlook excludes China data-center compute sales.

Longer-term evidence (2026-08-26): Release reports the Vera Rubin platform entering full production and describes the AI infrastructure product roadmap. A quantified ten-year company financial plan through 2036 was not verified.

Rivals: AMD; Intel. Annual report explicitly names both; compare product performance, software ecosystem and customer costs.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

AMD | Advanced Micro Devices

Semiconductors and chip equipment | One-page worksheet | All valuations below are illustrative

US | Nasdaq | USD earnings, US GAAP | FY0 ends 2025-12-27. Price USD 504.80 (30 Oct); market-cap proxy USD 824.1bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.16x / 0.08x / 0.09x
Value per share; bear at beta 1	USD 80.74; bear 39.47 (0.08x price); upside/downside -0.91
Reverse DCF; SBC	First-year growth 41.1%; reported SBC/revenue 4.7%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: 7; M: -2.65. Screening flags are not proof of manipulation.

Example assumptions: sales CAGR 13.6% repeated for five years, then fade to 3%; GAAP margin flat; tax 21.0%; WACC beta 1 9.9%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Risk: beta 2.49; volatility 69.9%; Sharpe 0.66; Treynor 0.18; alpha 28.3%. Worst price calendar year 2008: -71.2%; $6,000/loss = USD 8,427. Adjusted-return cap: USD 8,427. Historical caps are not recommended allocations.

Technical: -23.3% from 52-week high; USD price 504.80 vs 200-day mean 410.49; 1y price return 97.1%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): yes, 1.5%.

Guidance (2026-08-04): Q3 calendar 2026 revenue USD 13,000m +/-300m; non-GAAP gross margin approximately 56%.

Longer-term evidence (2025-11-11): Financial Analyst Day target for the next 3-5 years: revenue CAGR above 35%, non-GAAP operating margin above 35%, and non-GAAP EPS above USD 20. These are management targets, not a GAAP DCF forecast or a ten-year guarantee.

Rivals: NVIDIA; Intel. Annual report discusses both. Compare each in the relevant GPU/CPU market rather than assuming one company wins everywhere.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

AVGO | Broadcom

Semiconductors and chip equipment | One-page worksheet | All valuations below are illustrative

US | Nasdaq | USD earnings, US GAAP | FY0 ends 2025-11-02. Price USD 326.08 (30 Oct); market-cap proxy USD 1,556.6bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.84x / 0.60x / 0.66x
Value per share; bear at beta 1	USD 273.16; bear 72.48 (0.22x price); upside/downside -0.21
Reverse DCF; SBC	First-year growth 26.8%; reported SBC/revenue 11.8%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: 7; M: -2.10. Screening flags are not proof of manipulation.

Example assumptions: sales CAGR 24.4% repeated for five years, then fade to 3%; GAAP margin flat; tax 21.0%; WACC beta 1 9.7%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Risk: beta 1.49; volatility 42.0%; Sharpe 1.01; Treynor 0.28; alpha 32.0%. Worst price calendar year 2022: -16.0%; $6,000/loss = USD 37,565. Adjusted-return cap: USD 45,216. Historical caps are not recommended allocations.

Technical: -34.1% from 52-week high; USD price 326.08 vs 200-day mean 367.67; 1y price return -11.8%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): yes, 2.5%.

Guidance (2026-09-02): Q4 FY2026 revenue approximately USD 34,800m; non-GAAP operating margin 66%. These adjusted margins cannot replace GAAP operating margin without reconciling SBC and amortization.

Longer-term evidence (2026-09-02): Latest release describes custom AI accelerators, networking and infrastructure software. Verified quantitative horizon here is Q4 FY2026. A complete ten-year revenue/margin plan was not verified.

Rivals: NVIDIA; Marvell. Named in Broadcom FY2023 competitive disclosure. Historical evidence; confirm current segment overlap. Broadcom also has a large software business.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

QCOM | Qualcomm

Semiconductors and chip equipment | One-page worksheet | All valuations below are illustrative

US | Nasdaq | USD earnings, US GAAP | FY0 ends 2025-09-28. Price USD 164.79 (30 Oct); market-cap proxy USD 173.0bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.40x / 0.28x / 0.31x
Value per share; bear at beta 1	USD 66.72; bear 42.73 (0.26x price); upside/downside -0.80
Reverse DCF; SBC	First-year growth 12.5%; reported SBC/revenue 6.3%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: 6; M: -3.16. Screening flags are not proof of manipulation.

Example assumptions: sales CAGR 0.1% repeated for five years, then fade to 3%; GAAP margin flat; tax 56.2%; WACC beta 1 9.3%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Critical example limit: unusually high annual tax proxy and an elapsed, unreported first forecast year. Normalize only with evidence and rerun.

Risk: beta 1.71; volatility 45.1%; Sharpe 0.26; Treynor 0.07; alpha -0.4%. Worst price calendar year 2000: -53.3%; $6,000/loss = USD 11,250. Adjusted-return cap: USD 11,249. Historical caps are not recommended allocations.

Technical: -36.6% from 52-week high; USD price 164.79 vs 200-day mean 169.52; 1y price return -8.9%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): no; other ETF exposure remains possible.

Guidance (2026-07-29): Q4 FY2026 revenue USD 9,700-10,500m; GAAP diluted EPS USD 1.22-1.42; non-GAAP diluted EPS USD 2.05-2.25.

Longer-term evidence (2026-07-29): Management targets USD 40,000m non-handset revenue in FY2029 and above 60% non-handset growth in FY2027. Compare this newer plan with the older FY2025 annual base before choosing growth assumptions.

Rivals: MediaTek; Samsung. Named in the Q3 FY2026 10-Q. Samsung is also a customer/partner; segment distinctions matter.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

LRCX | Lam Research

Semiconductors and chip equipment | One-page worksheet | All valuations below are illustrative

US | Nasdaq | USD earnings, US GAAP | FY0 ends 2026-06-28. Price USD 239.11 (30 Oct); market-cap proxy USD 298.7bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.44x / 0.26x / 0.30x
Value per share; bear at beta 1	USD 105.62; bear 47.56 (0.20x price); upside/downside -0.70
Reverse DCF; SBC	First-year growth 21.8%; reported SBC/revenue 1.7%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: 8; M: -1.68. Screening flags are not proof of manipulation.

Example assumptions: sales CAGR 10.1% repeated for five years, then fade to 3%; GAAP margin flat; tax 12.1%; WACC beta 1 9.9%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Risk: beta 1.97; volatility 50.3%; Sharpe 0.76; Treynor 0.19; alpha 24.5%. Worst price calendar year 2000: -61.0%; $6,000/loss = USD 9,835. Adjusted-return cap: USD 9,835. Historical caps are not recommended allocations.

Technical: -45.5% from 52-week high; USD price 239.11 vs 200-day mean 284.24; 1y price return 51.9%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): no; other ETF exposure remains possible.

Guidance (2026-10-21): Quarter ending 27 December 2026: revenue USD 8,600m +/-400m; GAAP gross margin 52.5% +/-1 point; operating margin 40.5% +/-1 point; diluted EPS USD 2.33 +/-0.15.

Longer-term evidence (2026-10-21): Earnings release describes wafer-fabrication tools and installed-base support. A current, primary-source 2028 financial target was not verified here; secondary reports disagree on the range. No complete ten-year financial plan was verified.

Rivals: Applied Materials; Tokyo Electron. Annual report names both among principal equipment competitors.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

ASML | ASML

Semiconductors and chip equipment | One-page worksheet | All valuations below are illustrative

Netherlands | Euronext EUR / Nasdaq USD | EUR earnings, US GAAP | FY0 ends 2025-12-31. Price EUR 1,412.80 (30 Oct); market-cap proxy EUR 544.5bn. WInS eligibility MISSING.

Risk uses Nasdaq ASML USD 1,565.90; EUR valuation uses ASML.AS. USD beta in EUR DCF is a disclosed proxy.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.63x / 0.37x / 0.43x
Value per share; bear at beta 1	EUR 883.68; bear 316.20 (0.22x price); upside/downside -0.48
Reverse DCF; SBC	First-year growth 22.0%; reported SBC/revenue 0.6%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: NOT CHECKED; M: NOT CHECKED. Missing: LVGI; 1 F test unchecked.

Example assumptions: sales CAGR 15.6% repeated for five years, then fade to 3%; GAAP margin flat; tax 15.8%; WACC beta 1 8.3%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Risk: beta 1.80; volatility 42.8%; Sharpe 0.44; Treynor 0.10; alpha 6.0%. Worst price calendar year 2002: -51.0%; $6,000/loss = USD 11,772. Adjusted-return cap: USD 11,772. Historical caps are not recommended allocations.

Technical: -21.7% from 52-week high; USD price 1,565.90 vs 200-day mean 1,597.23; 1y price return 47.8%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): no; other ETF exposure remains possible.

Guidance (2026-10-14): Q4 2026 sales EUR 14,200-15,200m and gross margin around 57%; full-year sales EUR 44,000-45,000m and gross margin around 56%.

Longer-term evidence (2024-11-14): Investor Day presented a 2030 sales opportunity of EUR 44,000-60,000m with 56-60% gross margin under alternative scenarios. This older scenario is already near its low end in the newer FY2026 outlook. It is not guidance through 2036.

Rivals: Canon; Nikon. Annual report names both for DUV lithography. They are not equivalent commercial EUV competitors.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

RTX | RTX

Aerospace and defence | One-page worksheet | All valuations below are illustrative

US | NYSE | USD earnings, US GAAP | FY0 ends 2025-12-31. Price USD 199.80 (30 Oct); market-cap proxy USD 269.6bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.70x / 1.42x / 1.07x
Value per share; bear at beta 1	USD 139.84; bear 53.56 (0.27x price); upside/downside -0.41
Reverse DCF; SBC	First-year growth 14.0%; reported SBC/revenue 0.6%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: NOT CHECKED; M: NOT CHECKED. Missing: GMI; 1 F test unchecked.

Example assumptions: sales CAGR 9.7% repeated for five years, then fade to 3%; GAAP margin flat; tax 19.1%; WACC beta 1 9.3%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Risk: beta 0.31; volatility 24.2%; Sharpe 0.71; Treynor 0.56; alpha 15.0%. Worst price calendar year 1973: -46.6%; $6,000/loss = USD 12,867. Adjusted-return cap: USD 13,763. Historical caps are not recommended allocations.

Technical: -11.9% from 52-week high; USD price 199.80 vs 200-day mean 195.31; 1y price return 11.9%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): no; other ETF exposure remains possible.

Guidance (2026-10-20): FY2026 adjusted sales USD 97,000-97,500m; adjusted EPS USD 7.65-7.75; free cash flow USD 9,000-9,200m. Q3 adjusted EPS includes USD 0.25 from IEEPA recovery.

Longer-term evidence (2026-10-20): Reported backlog USD 340,000m, split between commercial aerospace and defence, supports investigation of delivery capacity and aftermarket demand. Backlog is not guaranteed future revenue or free cash flow.

Rivals: Lockheed Martin; Northrop Grumman. Defence comparison pair, supported by reciprocal competitive disclosures. These do not cover all commercial-engine or Collins product markets.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

LMT | Lockheed Martin

Aerospace and defence | One-page worksheet | All valuations below are illustrative

US | NYSE | USD earnings, US GAAP | FY0 ends 2025-12-31. Price USD 636.95 (30 Oct); market-cap proxy USD 147.1bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.64x / 1.84x / 1.14x
Value per share; bear at beta 1	USD 406.50; bear 185.99 (0.29x price); upside/downside -0.51
Reverse DCF; SBC	First-year growth 9.7%; reported SBC/revenue 0.4%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: NOT CHECKED; M: NOT CHECKED. Missing: GMI, SGAI; 1 F test unchecked.

Example assumptions: sales CAGR 4.4% repeated for five years, then fade to 3%; GAAP margin flat; tax 15.3%; WACC beta 1 9.3%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Risk: beta 0.08; volatility 29.5%; Sharpe 0.53; Treynor 2.03; alpha 15.1%. Worst price calendar year 1973: -64.4%; $6,000/loss = USD 9,319. Adjusted-return cap: USD 9,319. Historical caps are not recommended allocations.

Technical: -8.0% from 52-week high; USD price 636.95 vs 200-day mean 568.41; 1y price return 29.5%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): no; other ETF exposure remains possible.

Guidance (2026-10-22): FY2026 sales approximately USD 80,000-82,000m; diluted EPS USD 30.35-30.75; operating cash flow USD 10,000-10,200m; free cash flow USD 7,800-8,000m.

Longer-term evidence (2026-10-22): Reported backlog USD 292,000m and long-term missile production contracts. The release also warns that F-35 Lot 20 work exceeds current authorization/funding, which can hurt future cash flow.

Rivals: RTX; Northrop Grumman. Both appear in the annual competitor list; companies may also be suppliers or partners.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

GD | General Dynamics

Aerospace and defence | One-page worksheet | All valuations below are illustrative

US | NYSE | USD earnings, US GAAP | FY0 ends 2025-12-31. Price USD 372.59 (30 Oct); market-cap proxy USD 100.7bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.81x / 1.63x / 1.22x
Value per share; bear at beta 1	USD 300.74; bear 123.63 (0.33x price); upside/downside -0.29
Reverse DCF; SBC	First-year growth 13.0%; reported SBC/revenue 0.4%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: NOT CHECKED; M: NOT CHECKED. Missing: GMI, SGAI; 1 F test unchecked.

Example assumptions: sales CAGR 10.1% repeated for five years, then fade to 3%; GAAP margin flat; tax 17.5%; WACC beta 1 9.5%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Risk: beta 0.30; volatility 18.2%; Sharpe 0.65; Treynor 0.39; alpha 9.6%. Worst price calendar year 1990: -43.7%; $6,000/loss = USD 13,720. Adjusted-return cap: USD 14,280. Historical caps are not recommended allocations.

Technical: -6.9% from 52-week high; USD price 372.59 vs 200-day mean 354.87; 1y price return 8.0%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): no; other ETF exposure remains possible.

Guidance (2026-10-30): FY2026 revenue USD 56,500-56,800m; diluted EPS USD 17.20-17.30; operating margin around 10.5%. Verified in management remarks reproduced by The Motley Fool, a secondary transcript; the results release has no numeric forecast. Source is a secondary transcript.

Longer-term evidence (2026-10-30): Management discusses a goal of 200 Gulfstream aircraft deliveries in 2030 and expanding shipyard throughput. Source is a secondary transcript of management remarks; no complete ten-year financial forecast was verified.

Rivals: Lockheed Martin; Northrop Grumman. Defence comparison pair supported by reciprocal annual competitor lists. Separate Gulfstream and shipbuilding comparisons are still needed.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

NOC | Northrop Grumman

Aerospace and defence | One-page worksheet | All valuations below are illustrative

US | NYSE | USD earnings, US GAAP | FY0 ends 2025-12-31. Price USD 557.54 (30 Oct); market-cap proxy USD 79.2bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.70x / 3.35x / 1.52x
Value per share; bear at beta 1	USD 392.04; bear 170.64 (0.31x price); upside/downside -0.43
Reverse DCF; SBC	First-year growth 8.6%; reported SBC/revenue 0.3%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: NOT CHECKED; M: NOT CHECKED. Missing: GMI, DEPI, SGAI; 1 F test unchecked.

Example assumptions: sales CAGR 4.7% repeated for five years, then fade to 3%; GAAP margin flat; tax 17.5%; WACC beta 1 9.0%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Risk: beta -0.13; volatility 26.0%; Sharpe 0.37; Treynor n.a. (negative beta); alpha 10.6%. Worst price calendar year 2008: -42.7%; $6,000/loss = USD 14,043. Adjusted-return cap: USD 14,555. Historical caps are not recommended allocations.

Technical: -28.0% from 52-week high; USD price 557.54 vs 200-day mean 591.06; 1y price return -4.4%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): no; other ETF exposure remains possible.

Guidance (2026-10-20): FY2026 sales USD 43,750-44,250m; MTM-adjusted EPS USD 29.35-29.75; adjusted free cash flow USD 3,100-3,500m.

Longer-term evidence (2026-10-20): Management links capacity investment and backlog to its 2027 outlook. Annual risks include fixed-price development execution. No complete ten-year financial plan was verified.

Rivals: Lockheed Martin; RTX. Both named in the annual competitor list.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

HII | Huntington Ingalls Industries

Aerospace and defence | One-page worksheet | All valuations below are illustrative

US | NYSE | USD earnings, US GAAP | FY0 ends 2025-12-31. Price USD 316.88 (30 Oct); market-cap proxy USD 12.5bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.51x / 1.27x / 0.87x
Value per share; bear at beta 1	USD 160.52; bear 52.71 (0.17x price); upside/downside -0.59
Reverse DCF; SBC	First-year growth 12.7%; reported SBC/revenue 0.4%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: NOT CHECKED; M: NOT CHECKED. Missing: GMI, SGAI; 1 F test unchecked.

Example assumptions: sales CAGR 5.4% repeated for five years, then fade to 3%; GAAP margin flat; tax 22.1%; WACC beta 1 8.9%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Risk: beta 0.17; volatility 33.9%; Sharpe 0.38; Treynor 0.73; alpha 11.5%. Worst price calendar year 2020: -32.0%; $6,000/loss = USD 18,722. Adjusted-return cap: USD 19,699. Historical caps are not recommended allocations.

Technical: -31.1% from 52-week high; USD price 316.88 vs 200-day mean 335.65; 1y price return -1.6%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): no; other ETF exposure remains possible.

Guidance (2026-10-29): FY2026 shipbuilding revenue USD 10,500-10,700m; segment operating margin 6.0-6.5%; Mission Technologies revenue USD 3,000-3,200m. Full-year free cash flow USD 500-600m.

Longer-term evidence (2026-10-29): Management says awards have secured approximately 90% of expected shipbuilding revenue through 2030. Execution and cash conversion still matter; this is not 90% of total company revenue or guaranteed cash.

Rivals: General Dynamics; second segment-specific rival MISSING. General Dynamics Electric Boat is also a Virginia-class teammate. A second material rival needs verification by shipbuilding or Mission Technologies segment.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

ESLT | Elbit Systems

Aerospace and defence | One-page worksheet | All valuations below are illustrative

Israel | Nasdaq | USD earnings, US GAAP | FY0 ends 2025-12-31. Price USD 779.68 (30 Oct); market-cap proxy USD 36.2bn. WInS eligibility MISSING.

Model result	Value / status
M12, cross-check, 32 variants	Your kit run pending; add score, rank, shortlist and variants /32.
DCF value / price: beta 1, measured, adjusted	0.37x / 1.07x / 0.66x
Value per share; bear at beta 1	USD 290.08; bear 124.99 (0.16x price); upside/downside -0.75
Reverse DCF; SBC	First-year growth 27.9%; reported SBC/revenue 0.3%. SBC stays in GAAP EBIT.
Piotroski / Beneish	F: 8; M: -2.39. Screening flags are not proof of manipulation.

Example assumptions: sales CAGR 12.9% repeated for five years, then fade to 3%; GAAP margin flat; tax 4.7%; WACC beta 1 9.9%. Stress: -10%, 0%, half CAGR for years 3-5, then 2%; margin 25% lower. Full assumptions: methods page.

Risk: beta 0.09; volatility 32.5%; Sharpe 1.04; Treynor 3.90; alpha 33.3%. Worst price calendar year 2011: -23.0%; $6,000/loss = USD 26,130. Adjusted-return cap: USD 28,935. Historical caps are not recommended allocations.

Technical: -23.3% from 52-week high; USD price 779.68 vs 200-day mean 785.53; 1y price return 63.5%. Dip cause/permanence: MISSING. VOO top ten (30 Sep): no; other ETF exposure remains possible.

Guidance (2026-08-11): No quantitative forward revenue or EPS guidance identified in the latest available Q2 release. Reported Q2 revenue USD 2,300m and USD 32,000m backlog are historical facts, not guidance.

Longer-term evidence (2026-08-11): Management describes R&D and production-infrastructure investment to convert backlog into deliveries. No quantified ten-year revenue/EPS target or current annual revenue guidance was identified.

Rivals: Israel Aerospace Industries; Rafael. Annual report names both as competitors for Israeli government business. Other international markets have different competitors.

Sources: 10-K / 20-F | guidance | plan | rivals | price history | VOO Sep facts. Retrieved 2 Nov; source dates above. Accounting and history qualifications are in the workbook and missing-item register.

Your words: business; three positives / three negatives; competitor advantages; dip evidence.

Your edge and rank: what does price assume, what differs, and what would prove you wrong?

Your fit for Laura: horizon, portfolio overlap, proposed size and protection of payments.

How the calculations work

Method notes | Read before changing the inputs or interpreting a result

Dates and units

Research cutoff is 31 October 2026; market prices are the 30 October close. Fiscal years are company-specific. Amounts are millions in one reporting currency per company; EPS and prices are per share. ASML uses the US GAAP 20-F in EUR, not its different IFRS figures. ESLT uses the USD US GAAP 20-F. Outstanding shares set market cap; weighted-average diluted shares set DCF per-share value.

M12 and proposed rules

Each of eight measures is compared only with complete companies inside its group. Both groups meet the gate only with proposed R8 (unquantified finance leases use reported borrowings). Unquantified finance leases remain blank, not zero. The kit retains R1-R8 switches and cross-check options. R4 software capex applies only to Payments, so neither new group receives that adjustment. No student ranking or shortlist has been written.

Base and stress DCF examples

Historical revenue CAGR repeats for five years, then fades to 3% in year ten; reported GAAP operating margin, D&A and capex ratios remain flat. The common stress uses -10% revenue in year one, 0% in year two, half CAGR (floor zero) in years three-five, then fades to 2%; operating margin is 25% lower. These are classroom assumptions, not consensus or a company-specific bear thesis.

Discount rates and tax

USD 10-year Treasury: 4.94%; EUR current 10-year Bund: 3.342% with 9y9m remaining, both 30 October. Equity premium 5%, debt cost risk-free +0.5 percentage points, terminal return on new capital 15% are assumptions. Three beta cases use 1, measured, and 0.67 x measured + 0.33. The kit tax proxy is 1 - continuing net income / pretax income; negative AMD/AVGO proxies are replaced with a labelled 21% assumption. QCOM tax needs normalization.

Cash-flow conventions

GAAP EBIT already expenses SBC. Reported D&A is added back and extra sbc_pct is zero, avoiding a second deduction. Noncash working capital is a broad annual balance-sheet proxy. Cash includes corporate cash and identified liquid investments; unverified other non-operating assets are excluded. Annual cash/debt bridges count the whole first forecast year. QCOM estimated elapsed FY2026 cash is treated at time zero. The kit terminal formula includes D&A less capex plus terminal reinvestment; persistent acquired-intangible amortization may inflate AMD/AVGO values.

Risk and sizing

61 adjusted month-end prices from Oct 2021 to Oct 2026 give 60 monthly returns. Stock returns include dividends; ^GSPC is a price index, a disclosed benchmark mismatch. Beta is the regression slope; sample monthly volatility is annualized by sqrt(12). Sharpe, Treynor and alpha use annualized arithmetic returns and a 4.16% 3-month Treasury rate; CAGR is separate. NOC Treynor is n.a. because beta is negative. Worst full calendar years exclude partial IPO years and 2026. RTX/LMT include predecessors; AVGO begins in 2009. Both price and adjusted loss bases are supplied.

Red flags and interpretation

F-score and Beneish follow the kit; no score is declared complete if a required input is absent. No reported gross-profit subtotal means no invented GMI; G&A alone is not SG&A, nor is D&A pure depreciation. AMD red flags use continuing CFO, while M12 keeps reported total CFO. Pat must resolve this comparability choice. Proposed F >=4 and M <=-1.78 thresholds are screening conventions, not proof of safety or manipulation.

Rate sources: US Treasury, 30 Oct | Bundesbank, 30 Oct. The portfolio's locked 5.24% yield is unchanged. The kit's generated Markdown hardcodes USD/$; ASML results here are correctly labelled EUR.

Open items to resolve with Pat (1/3)

What is missing, what it changes, and how to acquire it

1. WInS eligibility for all 12

Importance: Critical before a simulated purchase

Effect: Approved exchanges, security availability and the USD 5 floor inside WInS. Public prices alone do not confirm eligibility.

Next route: Use the team WInS symbol search and record the exchange, quote date and eligibility. Browser access approval timed out during this research.

2. Comparable industry scorecard benchmarks

Importance: High for Round 1 score

Effect: Both industry totals remain MISSING. January 2026 US-industry reference values are older than the September/October S&P reference values, and there is no approved combined chip/equipment aggregation.

Next route: Get a single dated Finviz/FactSet/StockAnalysis industry and S&P extract. Specifically verify S&P operating margin, debt/equity, expected 5-year EPS growth and P/S; get each industry own 10-year forward-P/E average. Pat must approve any proxy or mixed-vintage comparison.

3. R8 sign-off

Importance: High for M12 release

Effect: Each group has 6/6 complete with proposed R8 and 3/6 without it. No strict group is rankable.

Next route: Pat can approve the disclosed reported-borrowings convention or request quantified finance leases from issuer relations. Do not turn unquantified amounts into zero.

Pat / Akshaj decision and date:

Open items to resolve with Pat (2/3)

What is missing, what it changes, and how to acquire it

4. ASML finance-lease maturity split

Importance: Medium; high for a complete red-flag score

Effect: EUR 16.5m FY0 / 16.9m FY1 total finance leases known, but current/noncurrent split is not. Piotroski leverage test and Beneish LVGI are not checked.

Next route: Ask ASML IR for the split or obtain a detailed lease schedule. Pat may explicitly approve a consistent borrowings-only red-flag sensitivity; do not silently substitute.

5. Gross-profit / SG&A / pure-depreciation disclosures

Importance: High for red-flag completeness

Effect: RTX, LMT, GD, NOC and HII lack a reported gross-profit subtotal. LMT lacks standalone SG&A; GD, NOC and HII disclose G&A only. NOC pure depreciation not verified. Incomplete scores remain NOT CHECKED.

Next route: Use issuer cost notes or IR clarification. The current kit prohibits constructing a Beneish GMI proxy. Any relaxed convention needs Pat approval.

6. DCF assumptions and stale annual balances

Importance: High for any valuation conclusion

Effect: All 12 examples need student-approved growth, margins, tax normalization, working capital and bear case. NVIDIA historical extrapolation is extreme; QCOM taxes and unreported elapsed fiscal year distort its example.

Next route: Replace examples with justified paths using dated guidance and recent statements. Normalize only documented one-offs and rerun all three betas. ASML also needs a currency-consistent beta check.

Missing pure interest income/expense also remains separately labelled in annual inputs where disclosure is combined. It does not block these GAAP-EBIT DCF examples, but it would matter for a pretax-to-EBIT reconstruction. ESLT identifies some interest components without certifying total pure interest.

Open items to resolve with Pat (3/3)

What is missing, what it changes, and how to acquire it

7. Exact cutoff shares, particularly ASML and ESLT

Importance: Medium

Effect: Market caps use latest reported outstanding counts, including December 2025 counts for the two foreign names. Buybacks/issuance can change valuation yields and weights.

Next route: Use the latest 6-K, share-repurchase disclosures or dated IR share register. Keep diluted weighted-average shares separate.

8. Three-year analyst revenue consensus as of cutoff

Importance: Medium; useful DCF cross-check

Effect: Not supplied as a verified historical snapshot. Company guidance and illustrative paths must not be called analyst consensus.

Next route: Obtain a dated Yahoo Analysis/StockAnalysis/FactSet export or WInS forecast table for FY1-FY3. A current November screen cannot be backdated to October 31.

9. Worst-year history and loss basis

Importance: High for position-size interpretation

Effect: Price and dividend-adjusted series give different caps. RTX/LMT contain predecessors; AVGO starts in 2009; vendor coverage is not a guarantee of full historical business risk.

Next route: Pat should choose the return basis and predecessor policy. Cross-check older calendar closes with Macrotrends or exchange records. Do not interpret the resulting size as a recommended purchase.

10. Qualitative research still owned by Akshaj

Importance: Required for submission

Effect: Edge, final rankings, strengths/weaknesses, dip cause and permanence, competitor advantages, and fit for Laura remain open. HII second material rival and several long-horizon targets are unverified.

Next route: Write evidence-based judgments, use segment-specific competitor filings, and bring drafts back for feedback. Complete the eight quiz answers. Record AI assistance in Works Cited.